

SECTION 6.2

1. Label the following statements as true or false.

(a) The Gram–Schmidt orthogonalization process allows us to construct an orthonormal set from an arbitrary set of vectors.

Answer : False. Consider the set $\{(1, 1), (2, 2), (3, 3)\}$ in $\mathbb{R}^2$.

(b) Every nonzero finite-dimensional inner product space has an orthonormal basis.

Answer : True. See the Theorem 6.5.

(c) The orthogonal complement of any set is a subspace.

Answer : True. We can prove it using the subspace test. Given a set W , we denote orthogonal complement of W by $W^\perp$. First, $W^\perp$ is nonempty. Indeed, 0 is in $W^\perp$ since $\langle 0, w \rangle = 0$ for any $w \in W$. Next, for any $x, y \in W^\perp$ and any $w \in W$,

$$\langle x + y, w \rangle = \langle x, w \rangle + \langle y, w \rangle = 0.$$

Finally, for any scalar $c \in F$, any $x \in W^\perp$, and any $w \in W$,

$$\langle cx, w \rangle = c\langle x, w \rangle = 0.$$

This shows that $W^\perp$ is a subspace.

(d) If $\{v_1, v_2, \dots, v_n\}$ is a basis for an inner product space V , then for any $x \in V$ the scalars $\langle x, v_i \rangle$ are the Fourier coefficients of x .

Answer : False. See the definition.

(e) An orthonormal basis must be an ordered basis.

Answer : True. See the definition.

(f) Every orthogonal set is linearly independent.

Answer : False. Note that any set of single vector is orthogonal. Thus, $\{0\}$ is orthogonal and linearly dependent.

(g) Every orthonormal set is linearly independent.

Answer : True. See Corollary 2 of Theorem 6.3.

2. In each part, apply the Gram–Schmidt process to the given subset S of the inner product space V to obtain an orthogonal basis for $\text{span}(S)$. Then normalize the vectors in this basis to obtain an orthonormal basis β for $\text{span}(S)$, and compute the Fourier coefficients of the given vector relative to β . Finally, use Theorem 6.5 to verify your result.

(f) Let $V = \mathbb{R}^4$, $S = \left\{ \begin{pmatrix} 1 \\ -2 \\ -1 \\ 3 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \\ 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \\ 2 \\ 8 \end{pmatrix} \right\}$, and $x = \begin{pmatrix} -1 \\ 2 \\ 1 \\ 1 \end{pmatrix}$.

Proof. The orthonormal basis β for $\text{span}(S)$ is

$$\left\{ \left(\frac{1}{\sqrt{15}}, -\frac{2}{\sqrt{15}}, -\frac{1}{\sqrt{15}}, \frac{3}{\sqrt{15}} \right), \left(\frac{2}{\sqrt{10}}, \frac{2}{\sqrt{10}}, \frac{1}{\sqrt{10}}, \frac{1}{\sqrt{10}} \right), \left(-\frac{4}{\sqrt{30}}, \frac{2}{\sqrt{30}}, \frac{1}{\sqrt{30}}, \frac{3}{\sqrt{30}} \right) \right\}.$$

The Fourier coefficients are

$$-\frac{3}{\sqrt{15}}, \frac{4}{\sqrt{10}}, \frac{12}{\sqrt{30}}.$$

□

(g) Let $V = M_{2 \times 2}(\mathbb{R})$, $S = \left\{ \begin{pmatrix} 3 & 5 \\ -1 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 9 \\ 5 & -1 \end{pmatrix}, \begin{pmatrix} 7 & -17 \\ 2 & -6 \end{pmatrix} \right\}$, and $A = \begin{pmatrix} -1 & 27 \\ -4 & 8 \end{pmatrix}$.

Proof. Recall that

$$\langle A, B \rangle = \text{tr}(B^* A) \text{ for } A, B \in V.$$

The orthonormal basis β for $\text{span}(S)$ is

$$\left\{ \begin{pmatrix} \frac{1}{2} & \frac{5}{6} \\ -\frac{1}{6} & \frac{1}{6} \end{pmatrix}, \begin{pmatrix} -\frac{\sqrt{2}}{3} & \frac{\sqrt{2}}{3} \\ \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{3\sqrt{2}} \\ \frac{\sqrt{2}}{3} & -\frac{\sqrt{2}}{3} \end{pmatrix} \right\}.$$

The Fourier coefficients are $24, 6\sqrt{2}, -9\sqrt{2}$.

□

(i) Let $V = \text{span}(S)$ with the inner product $\langle f, g \rangle = \int_0^\pi f(t)g(t) dt$, $S = \{\sin t, \cos t, 1, t\}$, and $h(t) = 2t + 1$.

Proof. The basis is

$$\beta = \left\{ \frac{\sqrt{2} \sin(t)}{\sqrt{\pi}}, \frac{\sqrt{2} \cos(t)}{\sqrt{\pi}}, \frac{\pi - 4 \sin(t)}{\sqrt{\pi^3 - 8\pi}}, \frac{8 \cos(t) + 2\pi t - \pi^2}{\sqrt{\frac{\pi^5}{3} - 32\pi}} \right\}.$$

And the Fourier coefficients are

$$\frac{\sqrt{2}(2\pi + 2)}{\sqrt{\pi}}, -\frac{4\sqrt{2}}{\sqrt{\pi}}, \frac{\pi^3 + \pi^2 - 8\pi - 8}{\sqrt{\pi^3 - 8\pi}}, \frac{\frac{\pi^4}{3} - 32}{\sqrt{\frac{\pi^5}{3} - 32\pi}}.$$

□

4. Let $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ i \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\} \subset \mathbb{C}^3$. Compute $S^\perp$.

Proof. If $(a, b, c) \in S^\perp$, then we have

$$\langle (a, b, c), (1, 0, i) \rangle = 0,$$

$$\langle (a, b, c), (1, 2, 1) \rangle = 0$$

This implies that

$$a - ci = 0 \text{ and } a + 2b + c = 0.$$

Therefore, $S^\perp = \text{span}\{(i, -\frac{(1+i)}{2}, 1)\}$. □

9. Let $W = \text{span}\left(\left\{\begin{pmatrix} i \\ 0 \\ 1 \end{pmatrix}\right\}\right) \subset \mathbb{C}^3$. Find orthonormal bases for W and $W^\perp$.

Proof. Note that W is 1-dimensional. Thus, we normalize the vector $(i, 0, 1)$ to obtain the orthonormal basis for W . The desired orthonormal basis for W is the set

$$\{(\frac{i}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}.$$

To find, $W^\perp$, let $(a, b, c) \in W^\perp$. Then, we have

$$\langle (a, b, c), ((i, 0, 1)) \rangle = 0.$$

This implies that $-ai + c = 0$. Thus, $W^\perp = \text{span}\{(1, 0, i), (0, 1, 0)\}$. Since this set is orthogonal, it suffice to normalize $W^\perp$. Therefore, the orthonormal basis for $W^\perp$ is

$$\{\frac{1}{\sqrt{2}}(1, 0, i), (0, 1, 0)\}.$$

□

10. Let W be a finite-dimensional subspace of an inner product space V . Prove that there exists a projection T on W along $W^\perp$ that satisfies $N(T) = W^\perp$. In addition, prove that $\|T(x)\| \leq \|x\|$ for all $x \in V$.

Hint: Use Theorem 6.6 and Exercise 10 of Section 6.1. (Projections are defined in the exercises of Section 2.1.)

Proof. We use Theorem 6.6. For any $x \in V$, there exist unique $u \in W, v \in W^\perp$ such that $x = u + v$. Thus, we define $T(x) = u$. Then, it is easy to see that $N(T) = W^\perp$.

Furthermore,

$$\begin{aligned}
 \|x\|^2 &= \langle x, x \rangle \\
 &= \langle u + v, u + v \rangle \\
 &= \langle u, u \rangle + \langle v, v \rangle \quad \text{since } \langle u, v \rangle = 0 \\
 &= \|u\|^2 + \|v\|^2 \\
 &= \|T(x)\|^2 + \|v\|^2
 \end{aligned}$$

This shows that $\|T(x)\| \leq \|x\|$. □

16.

- (a) **Bessel's Inequality.** Let V be an inner product space, and let $S = \{v_1, v_2, \dots, v_n\}$ be an orthonormal subset of V . Prove that for any $x \in V$ we have

$$\|x\|^2 \geq \sum_{i=1}^n |\langle x, v_i \rangle|^2.$$

Hint: Apply Theorem 6.6 to $x \in V$ and $W = \text{span}(S)$. Then use Exercise 10 of Section 6.1.

- (b) In the context of (a), prove that Bessel's inequality is an equality if and only if $x \in \text{span}(S)$.

Proof. (a) : Let W be the $\text{span}(S)$. Then for any $x \in V$, there exist $u \in W$ and $z \in W^\perp$ such that $x = u + z$. We have

$$\begin{aligned}
 \|x\|^2 &= \langle x, x \rangle \\
 &= \langle u + z, u + z \rangle \\
 &= \langle u, u \rangle + \langle z, z \rangle \quad (\text{since } \langle u, z \rangle = 0) \\
 &= \left\langle \sum_{i=1}^n \langle x, v_i \rangle v_i, \sum_{i=1}^n \langle x, v_i \rangle v_i \right\rangle + \|z\|^2 \\
 &= \sum_{i=1}^n |\langle x, v_i \rangle|^2 + \|z\|^2 \quad (\text{since } \langle v_i, v_j \rangle = 0 \text{ if } i \neq j).
 \end{aligned}$$

This shows that

$$\|x\|^2 \geq \sum_{i=1}^n |\langle x, v_i \rangle|^2.$$

(b) : By the equality in proof of (a), the inequality

$$\|x\|^2 \geq \sum_{i=1}^n |\langle x, v_i \rangle|^2.$$

is an equality if and only if $\|z\| = 0$. This implies that Bessel's inequality is an equality if and only if $x \in \text{span}(S)$. $\square$